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Department of Mathematics

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Math 208 Major Exam I Term 241

Original exam date: 1 October 2024. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. The solution of the system

$$x_1 - 2x_2 - 4x_3 + 8x_4 = 0$$

$$2x_1 + 3x_2 + 6x_3 + 9x_4 = 0$$

$$3x_1 + 5x_2 + 4x_3 + x_4 = 0$$

is the set of all scalar multiples of a vector $\mathbf{u}$, where $\mathbf{u} =$

- (a) $(-6, 5, 2, 1)$
- (b) $(-6, -5, -2, -1)$
- (c) $(-6, 5, -2, 1)$
- (d) $(-6, -5, -2, 1)$
- (e) $(6, 5, -2, 1)$

2. A general solution of $\frac{dy}{dx} = (x + y + 1)^2$ is

- (a) $y = -x - 1 + \sec(x + c)$
- (b) $y = -x + 1 + \tan(x + c)$
- (c) $y = x + 1 + \tan(x + c)$
- (d) $y = -x - 1 + \tan(x + c)$
- (e) $y = -x + 1 + \sec(x + c)$

3. The explicit particular solution of the initial-value problem $x^2 \frac{dy}{dx} + y = xy$, $y(1) = 2$ is $y(x) =$

- (a) $2xe^{x-1}$
- (b) $2x^2e^{x-1}$
- (c) $xe^{\frac{1}{x}+1}$
- (d) $2xe^{\frac{1}{x}-1}$
- (e) $2x^2e^{\frac{1}{x}-1}$

4. A general solution of the exact differential equation $(3x^2 + 2y^2) dx + (4xy + 6y^2) dy = 0$ is

- (a) $x^3 + 2xy^2 + 2y^3 = C$
- (b) $x^3 + xy^2 + y^3 = C$
- (c) $x^3 - 2xy^2 - 2y^3 = C$
- (d) $x^3 + 2xy^2 - 2y^3 = C$
- (e) $x^3 - 2xy^2 + 2y^3 = C$

5. A particle is moving in a straight line with acceleration $a(t) = 4(t + 3)^2$, initial position $x(0) = 1$ and initial velocity $v(0) = -1$. Then the position function $x(t)$ of the particle is given by

- (a) $x(t) = \frac{1}{4}(t + 3)^4 + 37t - 26$
- (b) $x(t) = \frac{1}{4}(t + 3)^4 - 37t - 26$
- (c) $x(t) = \frac{1}{3}(t + 3)^4 + 37t - 26$
- (d) $x(t) = \frac{1}{3}(t + 3)^4 - 30t - 26$
- (e) $x(t) = \frac{1}{3}(t + 3)^4 - 37t - 26$

6. By making a suitable substitution, the differential equation $x^4 y' = x^3 y - 5y^4$ can be transformed into the separable differential equation

- (a) $\frac{dv}{v^2} + \frac{5}{x} dx = 0$
- (b) $\frac{dv}{v^4} + \frac{5}{x} dx = 0$
- (c) $\frac{dv}{v^3} - \frac{5}{x} dx = 0$
- (d) $\frac{dv}{v^3} + \frac{5}{x} dx = 0$
- (e) $\frac{dv}{v^4} - \frac{5}{x} dx = 0$

7. Let $\mathbf{u} = (4, 5)$, $\mathbf{v} = (-2, 7)$, $\mathbf{w} = (8, 29)$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a - b =$

- (a) 3
- (b) 1
- (c) 2
- (d) 4
- (e) 0

8. The value of k for which the vectors $\mathbf{u} = (1, 4, 5)$, $\mathbf{v} = (4, 2, k)$, $\mathbf{w} = (-3, 3, -1)$ of $\mathbb{R}^3$ are linearly dependent is

- (a) $\frac{103}{15}$
- (b) $\frac{104}{15}$
- (c) $-\frac{96}{13}$
- (d) 0
- (e) $\frac{96}{13}$

9. By making a suitable substitution, the differential equation $xy' + 6y = 3xy^{4/3}$ can be transformed into the linear differential equation

- (a) $v' + \frac{2}{x}v = -1$
- (b) $v' - \frac{2}{x}v = 1$
- (c) $v' - \frac{2}{x}v = -1$
- (d) $v' - \frac{2}{x}v = x$
- (e) $v' + \frac{2}{x}v = x$

10. A general solution of the differential equation $x^2y'' + 2xy' = 12x^3$ is

- (a) $y(x) = x^3 + \frac{A}{x} + B$
- (b) $y(x) = x^3 - Ax^2 + B$
- (c) $y(x) = x^2 + \frac{A}{x} + B$
- (d) $y(x) = x^2 - \frac{A}{x} + B$
- (e) $y(x) = x^3 - \frac{A}{x} + B$

11. Find the constant A that makes the differential equation $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)dx + \left(\frac{Ax+1}{y^3}\right)dy = 0$ exact.

- (a) -2
- (b) 2
- (c) -3
- (d) 3
- (e) 0

12. Which one of the following statements is TRUE about the subset V of $\mathbb{R}^3$ defined by $V = \{(x_1, x_2, x_3) : x_2 = 1\}$?

- (a) V is not closed under addition but closed under multiplication by scalars
- (b) V is a subspace of $\mathbb{R}^3$
- (c) V is closed under addition
- (d) V is closed under addition and not closed under multiplication by scalars
- (e) V is not closed under addition and not closed under multiplication by scalars

13. The general solution of the linear differential equation $2xy' - 3y = 9x^3$ is given by

- (a) $y(x) = 2x^3 + cx^{5/2}$
- (b) $y(x) = x^2 + cx^{3/2}$
- (c) $y(x) = 3x^3 + cx^{3/2}$
- (d) $y(x) = 3x^2 + cx^{-3/2}$
- (e) $y(x) = x^3 + cx^{2/3}$

14. The sum of all values of r such that $y = e^{rx}$ is a solution of the differential equation $3y'' + 3y' - 4y = 0$ is

- (a) 0
- (b) -1
- (c) 1
- (d) $-\frac{3}{2}$
- (e) $\frac{1}{2}$

15. If a certain substance cools from 100°C to 60°C in 10 minutes when it is taken outside where the air temperature is 20°C , then the temperature of the substance 40 minutes after it is taken outside is

- (a) 35°C
- (b) 25°C
- (c) 10°C
- (d) 15°C
- (e) 40°C

Answer Key

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Q	Ans	Brief explanation
1	(c)	Row-reduce, $x_4 = t$: $x_3 = -2t$, $x_2 = 5t$, $x_1 = -6t$.
2	(d)	$v = x + y + 1$ gives $v' = 1 + v^2$, so $v = \tan(x + c)$ and $y = -x - 1 + \tan(x + c)$.
3	(d)	Separable: $\frac{dy}{y} = (\frac{1}{x} - \frac{1}{x^2})dx$ gives $y = Kxe^{1/x}$; $y(1) = 2$ gives $K = 2/e$.
4	(a)	$M_y = N_x = 4y$. $F = \int M dx = x^3 + 2xy^2 + g(y)$ and $g' = 6y^2$ gives $g = 2y^3$.
5	(e)	$v = \frac{4}{3}(t+3)^3 - 37$ (from $v(0) = -1$); $x = \frac{1}{3}(t+3)^4 - 37t - 26$ (from $x(0) = 1$).
6	(b)	Homogeneous: $v = y/x$ gives $v + xv' = v - 5v^4$, so $\frac{dv}{v^4} + \frac{5}{x}dx = 0$.
7	(b)	$4a - 2b = 8$, $5a + 7b = 29$ give $a = 3$, $b = 2$, so $a - b = 1$.
8	(b)	Dependent iff $\det = 0$; expanding gives $104 - 15k = 0$, so $k = \frac{104}{15}$.
9	(c)	Bernoulli, $n = \frac{4}{3}$, $v = y^{-1/3}$: $-3v' + \frac{6}{x}v = 3$, i.e. $v' - \frac{2}{x}v = -1$.
10	(e)	y missing: $p = y'$ gives $(x^2p)' = 12x^3$, $p = 3x^2 + Ax^{-2}$, $y = x^3 - \frac{A}{x} + B$.
11	(a)	Exact iff $M_y = N_x$: $-\frac{2}{y^3} = \frac{A}{y^3}$, so $A = -2$.
12	(e)	$(a, 1, b) + (c, 1, d)$ and $2(a, 1, b)$ both have middle entry 2, so both closures fail.
13	(c)	Standard form $y' - \frac{3}{2x}y = \frac{9}{2}x^2$, $\rho = x^{-3/2}$, so $x^{-3/2}y = 3x^{3/2} + c$.
14	(b)	Substituting e^{rx} gives $3r^2 + 3r - 4 = 0$; the roots sum to $-3/3 = -1$.
15	(b)	$T = 20 + 80e^{-kt}$, $e^{-10k} = \frac{1}{2}$, so $T(40) = 20 + 80 \cdot \frac{1}{16} = 25$.